



COMPUTER NETWORKS

LAB 07



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IFZA RIZWAN
SAP 64281

Task 1:

Send a ping command for both HUB and SWITCH network using command prompt in packet tracer and notice the results in simulation mode.

Scenario:

Create simplest hub-based and switched-based networks, ideal for studying the different processes between hubs and switches. Add generic PCs and 2950-24 switches for one topology and generic hub with PCs the other topology.

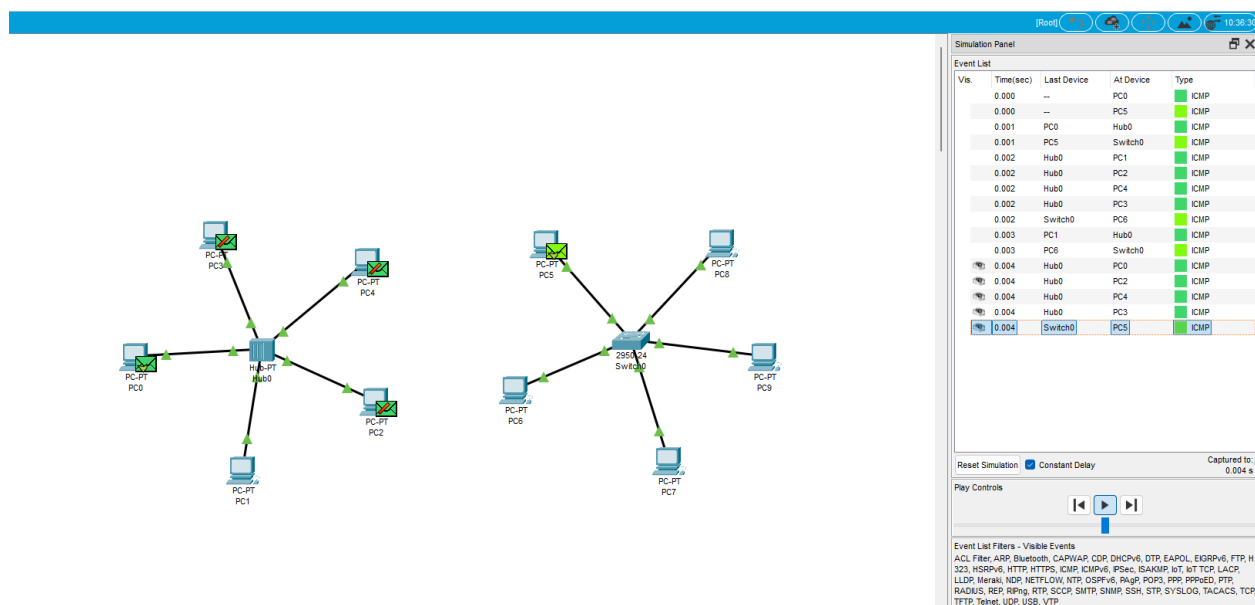
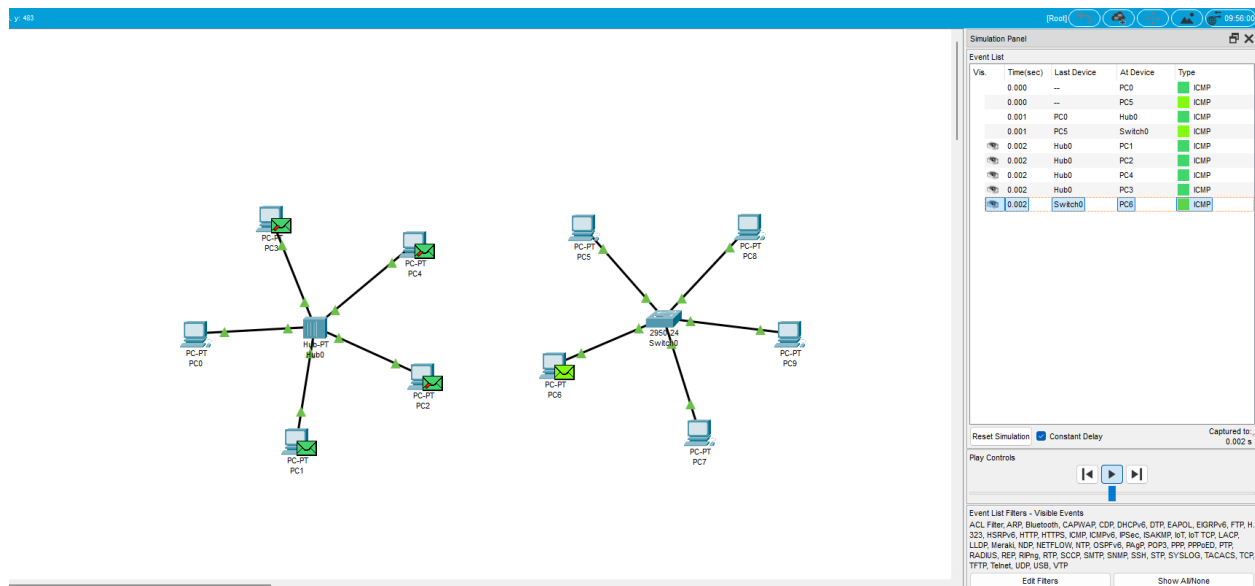
Set the IP addresses ranging from 192.168.1.1 to 192.168.1.5 for one topology and 192.168.2.1 to 192.168.2.5 for another topology. All should have subnet masks 255.255.255.0.

Reflect:

Go to simulation mode. In event list filters, enable only ICMP and ARP. Using the “Simple PDU”, issue a ping from PC 0 to PC 1. Play the simulation. Pay close attention to how the hub processes the ICMP and ARP packets. After that, once again, use” Add simple PDU to issue a ping from PC1 to PC 0. Play the simulation again. How has the behavior of the hub changed from the first and second attempts, if at all?

Still in simulation mode, in the event List Filters, enable only ICMP and ARPO. Using the Add simple PDU issue a ping from PC 3 to PC 4. Play the simulation. Pay close attention to how the switch processes the ICMP and ARP packets. Now check for the behavior for the Switch.

In what ways did the switch processes the packets similarly or differently from Hub between the first and second ping attempts?



Task 2:

For the switched based network, set the new IP address and subnet mask for the PCs as follows.

PC	IP ADDRESSES	SUBNET MASK
PC0	192.168.1.36	255.255.255.224
PC1	192.168.1.37	255.255.255.224
PC2	192.168.1.38	255.255.255.224
PC3	192.168.1.91	255.255.255.224

Tell about the communication. Communicated or not? If yes why.

[Root]
03.4

PC-PT PC0 PC-PT PC3
Switch0
PC-PT PC1 PC-PT PC2

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP

Reset Simulation ☒ Constant Delay Captured: 0.00

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTI, 323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PA-p, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, ...

PC-PT PC0 PC-PT PC3
Switch0
PC-PT PC1 PC-PT PC2

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP

Reset Simulation ☒ Constant Delay Captured to: 0.004 s

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H, 323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PA-p, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, ...

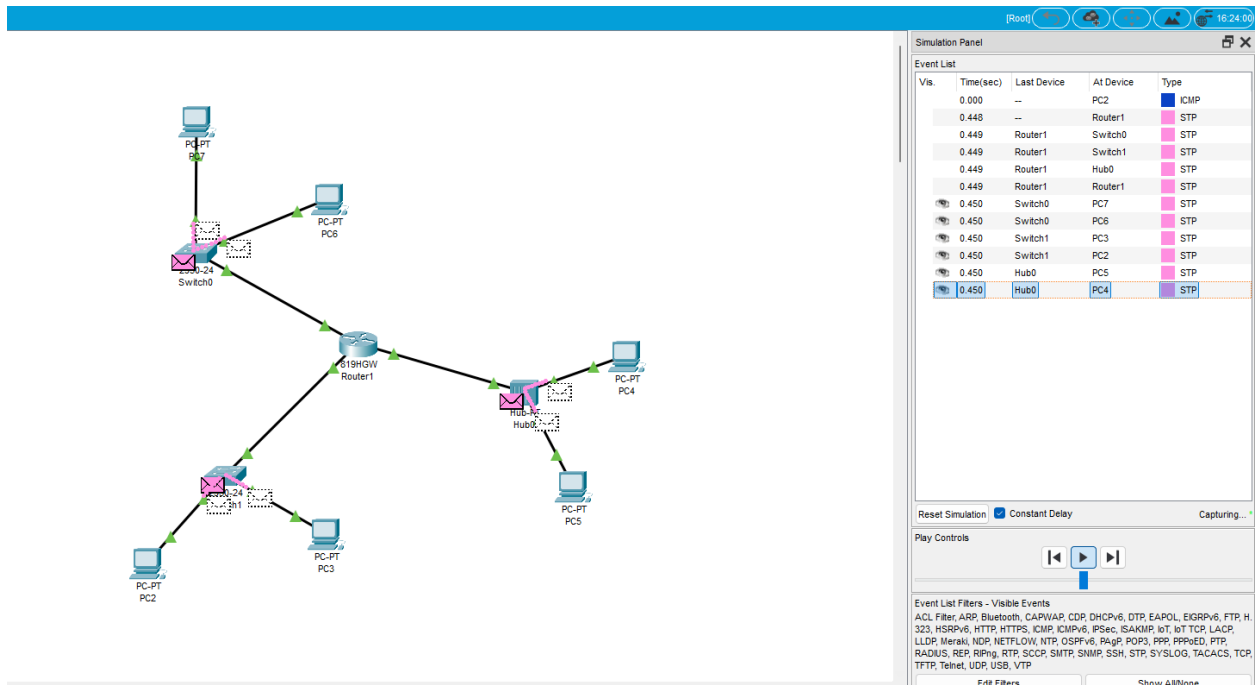
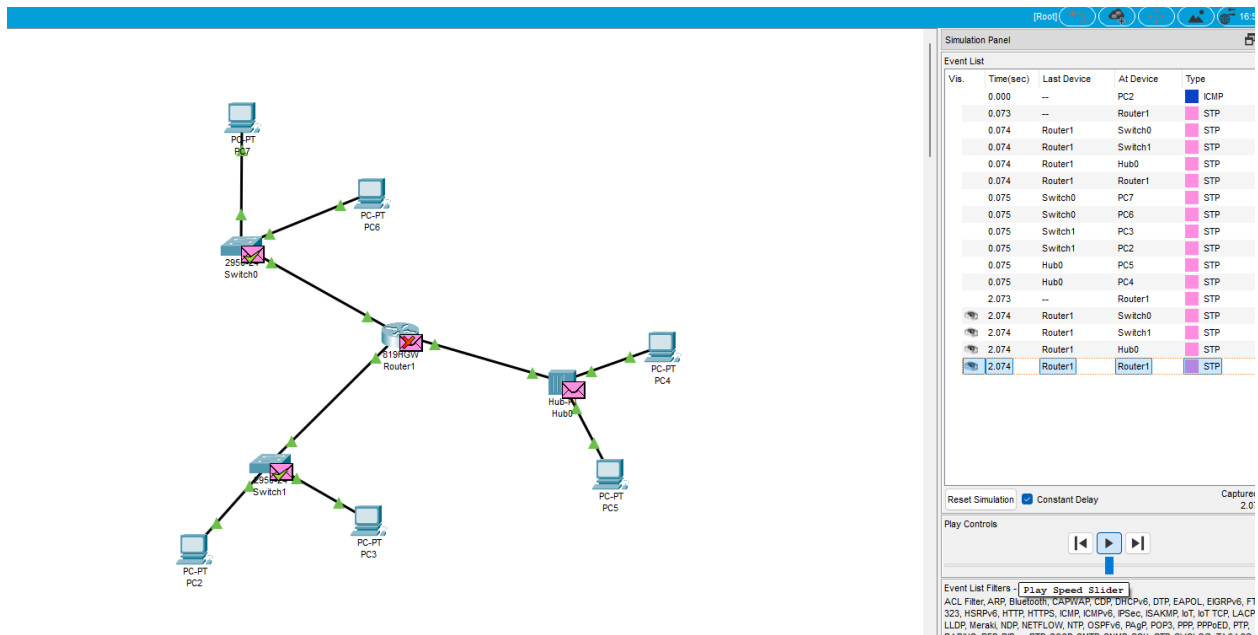
PC 0 and PC 1 communicate with each other because they are in same subnet range.

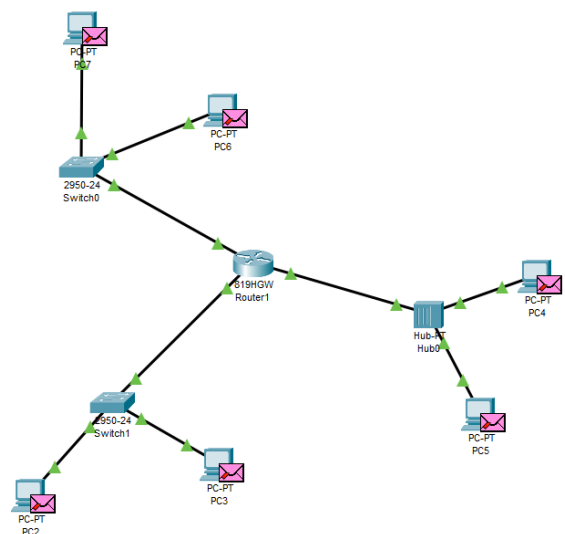
The top screenshot shows a star topology with a central switch labeled '2950-24 Switch0' connected to four PCs: PC0, PC1, PC2, and PC3. All connections are active. The corresponding Event List window shows a sequence of events from 0.000 to 3.006 seconds, including ICMP and STP messages between the switch and the PCs. The bottom screenshot shows the same topology, but PC0 and PC3 are disconnected, indicated by red 'X' marks on their connection lines. The Event List window for this state shows events continuing up to 7.824 seconds, including a DTP event at 7.824 seconds.

PC 0 and PC 3 does not communicate with each other because they are not in same subnet range.

Task 3:

Create a star topology using a class A addressing scheme. Use Router to connect the end devices. Explain the behavior of the ping on each device. Use one router with 4 networks, 2 switches and 1 hub.





Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC2	ICMP
	0.858	--	Router1	STP
	0.857	Router1	Switch0	STP
	0.857	Router1	Switch1	STP
	0.857	Router1	Hub0	STP
	0.857	Router1	Router1	STP
	0.858	Switch0	PC7	STP
	0.858	Switch0	PC6	STP
	0.858	Switch1	PC3	STP
	0.858	Switch1	PC2	STP
	0.858	Hub0	PC5	STP
	0.858	Hub0	PC4	STP

Reset Simulation ☒ Constant Delay Captured to: 0.858

Play Controls: ⏮ ▶ ⏭

Event List Filters - Visible Events
 ACL Filter: ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, 323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TFTP, Telnet, UDP, USB, VTP

(Root)
18:25:30

```

graph TD
    Router1[619HGW Router1] --- Switch0[2950-24 Switch0]
    Router1 --- Switch1[2950-24 Switch1]
    Router1 --- Hub0[Hub0]
    Switch0 --- PC7[PC-PT PC7]
    Switch0 --- PC6[PC-PT PC6]
    Switch1 --- PC2[PC-PT PC2]
    Switch1 --- PC3[PC-PT PC3]
    Hub0 --- PC4[PC-PT PC4]
    Hub0 --- PC5[PC-PT PC5]
  
```

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	2.042	Router1	Switch0	STP
	2.042	Router1	Switch1	STP
	2.042	Router1	Hub0	STP
	2.042	Router1	Router1	STP
	2.043	Switch0	PC7	STP
	2.043	Switch0	PC6	STP
	2.043	Switch1	PC3	STP
	2.043	Switch1	PC2	STP
	2.043	Hub0	PC5	STP
	2.043	Hub0	PC4	STP
	4.044	--	Router1	STP
	4.045	Router1	Switch0	STP
	4.045	Router1	Switch1	STP
	4.045	Router1	Hub0	STP
	4.045	Router1	Router1	STP
	4.046	Switch0	PC7	STP
	4.046	Switch0	PC6	STP
	4.046	Switch1	PC3	STP
	4.046	Switch1	PC2	STP
	4.046	Hub0	PC5	STP
	4.046	Hub0	PC4	STP
	5.179	--	Switch1	DTP
	5.180	Switch1	PC2	DTP

Reset Simulation ☒ Constant Delay Captured to: 5.180 s

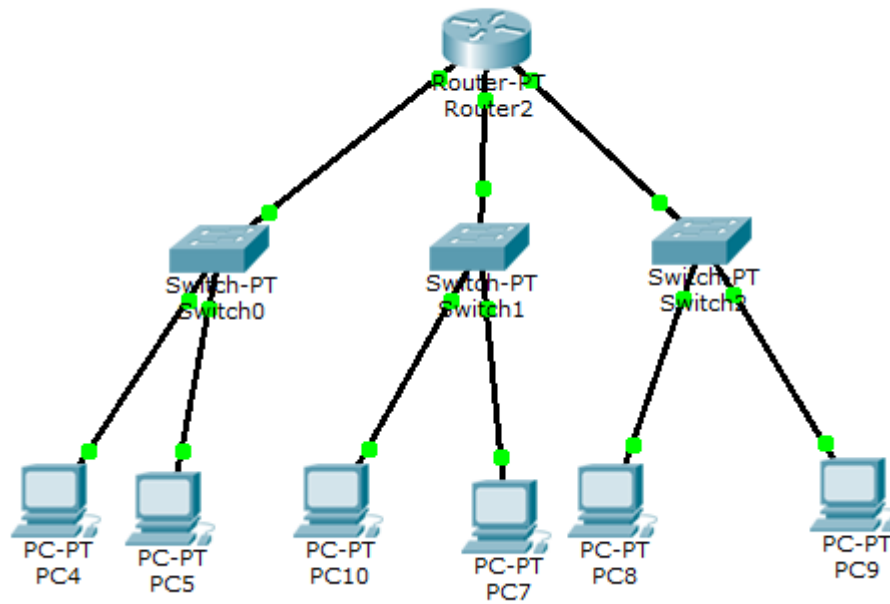
Play Controls: ⏮ ▶ ⏭

Event List Filters - Visible Events
 ACL Filter: ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H. 323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

[Edit Filters](#) [Show AllNone](#)

Task 4:

Create the topology shown in the below diagram



Send packets and ping commands and provide results with explanation.

Task 5:

What are ICMP and ARP in cisco packet tracer?

ARP:

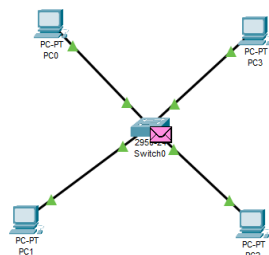
- Maps IP to MAC.
- Used when one device wants to learn MAC of another.

ICMP:

- Used by **ping** (Echo Request/Reply).

To Observe ARP and ICMP:

1. **Switch to Simulation Mode.**
2. **Event Filters** → Enable only ARP and ICMP.
3. **Use Simple PDU tool** to ping one PC to another.
4. **Click Capture/Forward** to step through:
 - First: ARP Request & Reply
 - Then: ICMP Echo Request & Reply



Vis.	Time(sec)	Last Device	At Device	Type
☐	0.000	--	PC0	ICMP
☐	0.000	--	PC0	ICMP
☐	0.001	PC0	Switch0	ICMP
☐	0.002	Switch0	PC1	ICMP
☐	0.003	PC1	Switch0	ICMP
☐	0.004	Switch0	PC0	ICMP
☒	0.183	--	Switch0	STP

This is an example from above of the tasks that how ICMP and ARP works.

THE END